KOSEI FUJIKI

Ph.D. Student in Theoretical Particle Physics
Yukawa Institute for Theoretical Physics (YITP), Kyoto University
kosei.fujiki@yukawa.kyoto-u.ac.jp | fujikikosei.github.io | ORCID 0009-0003-8847-105X | INSPIRE-HEP

Google Scholar | arXiv author search | X: @fffff_phys

RESEARCH PROFILE

Theoretical particle physicist working toward a quantum theory of gravity through holography. Research focuses on quantum-information aspects of AdS/CFT, especially multipartite quantum correlations and their geometric duals, together with extensions of holography to dS/CFT and AdS/BCFT.

Research interests: quantum gravity; holography; AdS/CFT; dS/CFT; AdS/BCFT; multipartite entanglement; black-hole information; gravitational path integrals; non-Hermitian CFT; pseudo entropy; complexified geometry; brane cosmology.

EDUCATION

2026-present — **Ph.D. Program in Physics**
Yukawa Institute for Theoretical Physics, Kyoto University | Supervisor: Prof. Tadashi Takayanagi

2024-2026 — **M.Sc. in Physics**
Yukawa Institute for Theoretical Physics, Kyoto University | Supervisor: Prof. Tadashi Takayanagi

2020-2024 — **B.Sc. in Physics**
Faculty of Science, Kyoto University

PUBLICATIONS

1. **Detecting One-Dimensional Bosonic SPT Phases via Twisted Entropic Order Parameter**
Kosei Fujiki, Tsubasa Oishi, and Soichiro Shimamori
arXiv:2609.11089 [hep-th, cond-mat.str-el, quant-ph] (2026). arXiv

2. **Effective Dynamics of Inflationary End-of-the-World Branes in AdS_3**
Kosei Fujiki, Michitaka Kohara, Jason Kristiano, Yu-ki Suzuki, and Tadashi Takayanagi
arXiv:2609.11643 [hep-th, gr-qc] (2026). arXiv

3. **De Sitter Holography from a Lorentzian Torus**
Kosei Fujiki, Michitaka Kohara, Javier Moreno, Kotaro Shinmyo, and Tadashi Takayanagi
arXiv:2608.01729 [hep-th, gr-qc] (2026). arXiv

4. **The Entanglement Wedge Polygon**
Kosei Fujiki, Jonathan Harper, Tadashi Takayanagi, and Nicolò Zenoni
arXiv:2606.21081 [hep-th] (2026). arXiv

5. **Multi-Entropy in Heavy Local Quenches**
Kosei Fujiki and Kenya Tasuki
arXiv:2606.12526 [hep-th, cond-mat.stat-mech, quant-ph] (2026). arXiv

6. **Entropic Interpretation of Einstein Equation in dS/CFT**
Kosei Fujiki, Michitaka Kohara, Kotaro Shinmyo, Yu-ki Suzuki, and Tadashi Takayanagi
Journal of High Energy Physics 04 (2026) 042; arXiv:2511.07915 [hep-th]. arXiv

7. **Brane Cosmology from AdS/BCFT**
Kosei Fujiki, Hiroki Kanda, Michitaka Kohara, and Tadashi Takayanagi
Journal of High Energy Physics 03 (2025) 135; arXiv:2501.05036 [hep-th]. arXiv

AWARDS AND FUNDING

2026 **Division of Graduate Studies Donor Designated Scholarship** - Kyoto University Division of Graduate Studies, Fujitsu Limited
2026 **JSPS Research Fellowship for Young Scientists (DC1)** - Japan Society for the Promotion of Science
2024 **Poster Presentation Award** - 2024 Summer School for Young Researchers

TALKS AND PRESENTATIONS

Sep 2026 **Multi-Entropy after a Quantum Local Quench in CFT and Holography** - 81st Annual Meeting of the Physical Society of Japan, Tokyo

Aug 2026 **Entanglement Wedge Polygon: A Geometric Measure of Multipartite Correlations** - Strings and Fields 2026, Kyoto

Aug 2026 **Multi-Entropy after a Quantum Local Quench in CFT and Holography** - New Developments in Quantum Foundations and Quantum Information, Kyoto

Jul 2026 **[Selected]** **Multi-Entropy in Heavy Local Quenches** - Strings 2026 Gong Show, Shanghai

Jul 2026 **[Poster]** **Multi-Entropy after a Quantum Local Quench in CFT and Holography** - Strings 2026, Shanghai

Jun 2026 **Entanglement Wedge Polygon: A Geometric Measure of Multipartite Correlations** - Quantum Information in Quantum Gravity 2026, Beijing

Jun 2026 **[Selected]** **Multi-Entropy in Heavy Local Quenches** - Holo-Asia 2026 Gong Show, Jeju Island

Jun 2026 **The Einstein Equation and the First-Law-like Relation of Entanglement Entropy in dS/CFT** - de Sitter Holography Meets Non-Hermitian Quantum Matter, Wako

Jun 2026 **Multi-Entropy after a Quantum Local Quench in CFT and Holography** - Quantum Thermalization, Hydrodynamics and Gravity, Kyoto

May 2026 **[Invited]** **The Einstein Equation and the First Law in dS/CFT** - Institute of Science Tokyo Seminar, Tokyo

May 2026 **[Invited]** **Brane Cosmology and Inflation Models with a Localized Scalar Field** - Nihon University Particle Physics Seminar, online

May 2026 **Multi-Entropy after a Quantum Local Quench in CFT and Holography** - Holographic Universe 2026, Kyoto

Mar 2026 **The Einstein Equation and the First Law of Entanglement Entropy in dS/CFT** - 1st Symposium of the Center for Education and Research in Optical Quantum Sensing, Kyoto

Mar 2026 **Brane Cosmology and Inflation Models with a Localized Scalar Field** - Physical Society of Japan Spring Meeting, online

Dec 2025 **Analysis of dS/CFT from AdS/BCFT with a Localized Scalar Field** - Extreme Universe Fifth Annual Meeting, Matsuyama

Oct 2025 **[Invited]** **Analysis of dS/CFT and Cosmological Models from AdS/BCFT with a Localized Scalar Field** - KEK Theory Seminar, Tsukuba

Oct 2025 **Einstein Equation and First-Law-like Relation of Entanglement Entropy in dS/CFT** - Extreme Universe 2025, Kyoto

Sep 2025 **Einstein Equation and the First Law in dS/CFT** - 80th Annual Meeting of the Physical Society of Japan, Hiroshima

Aug 2025 **Progress in Two-Dimensional Quantum Gravity (JT Gravity) from Black-Hole Chaos** - 2025 Summer School for Young Researchers, Kyoto

Jul 2025 **dS/CFT from AdS/BCFT with a Localized Scalar Field** - Strings and Fields 2025, Kyoto

Jun 2025 **Einstein Equation and First-Law-like Relation of Entanglement Entropy in dS/CFT** - 4th Young Researchers' Workshop of the Extreme Universe Collaboration, Aichi

May 2025 **dS/CFT from AdS/BCFT with a Localized Scalar Field** - Black Holes, Quantum Chaos and Quantum Information, Kyoto

Apr 2025 **Restriction on Non-Minimal RT Surfaces from Holographic Scattering** - Journal Club at YITP, Kyoto

Apr 2025 **Analysis of dS/CFT and Cosmological Models with an EOW-Brane-Localized Scalar Field** - M1 Presentation, Kyoto University

Feb 2025 **The Connected Wedge Theorem from Quantum Tasks** - Entanglement Meeting, Kyoto

Feb 2025 **dS/CFT from AdS/BCFT with a Localized Scalar Field** - Quantum Gravity and Information in an Expanding Universe, Kyoto

Feb 2025 **Analysis of dS/CFT from AdS/BCFT with a Localized Scalar Field** - Theoretical Physics Students Seminar 2025, online

Jan 2025 **dS/CFT from AdS/BCFT with a Localized Scalar Field** - Gravity 2025, Kyoto

Jan 2025 **dS/CFT from AdS/BCFT with a Localized Scalar Field** - Recent Developments in Black Holes and Quantum Gravity, Kyoto

Sep 2024 **The Page Curve of Hawking Radiation** - 3rd Young Researchers' Workshop of the Extreme Universe Collaboration, Otaru

Aug 2024 **The Page Curve of Hawking Radiation** - 2024 Summer School for Young Researchers, Tokyo

OUTREACH

Jun 2026 JSPS Science Dialogue Assistant, Wakasa High School, Fukui.

Office: Room H301, North Building, YITP, Kyoto University, Kitashirakawa Oiwakecho, Sakyo-ku, Kyoto 606-8502, Japan